

Subsurface Conditions Part I:

Basic Concerns/Design Information/Division of Responsibilities

National Council of Architectural Registration Boards

1401 H Street NW, Suite 500

Washington, DC 20005

Tel: 202-879-0520

Fax: 202-783-0290

Web: <http://www.ncarb.org>

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Preface

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Course Description and Learning Objectives

Course Description:

Soil and subsurface water conditions affect the design and performance of building projects, therefore, knowledge about subsurface conditions is critical. This mini-monograph discusses design tasks related to subgrade conditions, coordination of design professionals and division of responsibility, and the determination of what information is required and how to obtain it.

This course is part of a four-part series that examines soil mechanics as it relates to building construction and performance. Each section can be taken as an individual course.

Learning Objectives:

At the end of this course participants will be able to:

- create and coordinate an appropriate team of subsurface condition experts to obtain, in a timely manner, the necessary information for the design of safe and healthy buildings
- acquire the necessary information to develop site designs that avoid creating long-term environmental problems such as erosion, sinking soils, and lowering of local water tables
- identify potential subsurface issues at the earliest possible stage to develop solutions that minimize their impact on natural conditions, and
- monitor site exploration and construction activities to determine the degree of accuracy or completeness of early subsurface information in order to propose any design changes necessary to avoid dangerous or unhealthy post-occupancy conditions and expensive remedial actions.

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Foreword

This program is intended to help architects meet mandatory continuing education and professional development requirements of state registration boards. It is also accepted by the American Institute of Architects to meet its continuing education requirements for maintenance of membership in the Institute.

This mini-monograph has been created from the *Subsurface Conditions* monograph. While it has been updated by NCARB to remain timely, it has not been extensively rewritten. Thus, in most jurisdictions, an individual who has already read the first edition cannot read the second edition for continuing education credit.

Subsurface Conditions is intended to assist architects by presenting information, which bridges a gap between knowledge commonly gained about soils science in a university internship setting and knowledge believed necessary for a practicing architect to converse competently on the topic. Although technology has changed since 1995, the Monograph Program felt that the basics involved in understanding subsurface materials and construction methods covered in the original text are still valid today in the 21st century.

Foreword

NCARB's Education Committee selects the topics and reviews the content for each new monograph. The committee also reviews titles currently in print for timeliness and applicability of general concepts. Where appropriate, the monographs are revised and republished.

NCARB welcomes your comments and questions about this mini-monograph and about the Monograph Program in general. Please let us know what other topics you would like to see addressed. Thank you for your participation.

About the Author

James Ambrose is an architect and structural engineer and has been the author, co-author, contributor, or editor of more than 50 books on architectural technology and design. In his 35-year career in college teaching, he has served on the faculties of architecture schools at the Universities of Illinois, Wisconsin, and Southern California.

As a working design professional, Ambrose has conducted practices as a private consultant and as a member of partnerships in Urbana, IL, Chicago, Los Angeles, and Milwaukee. He has also worked as a design structural engineer for Skidmore, Owings and Merrill in Chicago and for Albert C. Martin in Los Angeles.

About the Author

As a faculty member at professional schools of architecture, Ambrose has taught courses in architectural technology, as well as design studios, service courses for art and engineering students, and short courses for working professionals. He has also served as an academic administrator for programs in Milwaukee and Los Angeles.

His recent writing projects include preparation of the eighth edition of the late Harry Parker's *Simplified Engineering for Architects and Builders*, a standard for structures for architects that is still used widely today as a test and as a study reference for the Architect Registration Examination®. Other recent titles include *Building Structures*, *Design of Building Trusses*, *Construction Revisited*, and *Simplified Design of Building Structures*.

In the topic areas, soils and foundations, Ambrose has written *Simplified Design of Building Foundations*, 2nd edition; *Simplified Site Engineering*, 2nd edition; *Simplified Site Design* with Peter Brandow; and *Building Construction: Site and Below-Grade Systems*.

About the Contributors

Several consultants provided expert assistance during the development of this monograph: their experience in the foundation and subgrade work was invaluable to the author.

Dov Kaminetzky is a partner in the New York City consulting firm of Feld, Kaminetzky and Cohen. The firm is engaged in the general practice of structural engineering and is especially known for its work in forensic engineering, including many cases of foundation failures. Kaminetzky is the author of *Design and Construction Failures: Lessons from Forensic Engineering*.

Luther Boudra is a geotechnical consultant with many years of experience in engineering investigation and design for foundations and general subgrade problems. Boudra is an assistant vice president of the environmental engineering consulting company, Law Engineering, Inc., for whom he works as chief engineer in the Birmingham, AL office.

Introduction

Architecture, as we generally know it, exists above the ground. Only incidentally are building spaces sometimes extended below grade, usually for utilitarian purposes. Fully subterranean buildings are sometimes built, but very rarely.

However, every building rests on, and in, the ground. Every building has a foundation system, underground hookups to services, and a developed site that uses the ground surface. The horizontal and vertical placement of a building relates to the geography and geology of the Earth's surface materials. For a building's design, and most especially for the development of its construction, the surface and subsurface ground conditions must be discovered in detail, and all problems that may arise from these conditions must be evaluated fully.

As in many areas of architectural design and practice, there are more considerations to deal with in the area of subsurface conditions than most architects can afford the time and effort to handle adequately. The services of other professionals and specialists who are trained and experienced in dealing with subsurface conditions are often required. Architects must understand the potential problems and the dimensions of the necessary work and must make judgments about what can be done about them. What needs doing? Who will be needed to do it? When does it need to be done to avoid holding up the progress of the design or construction work?

Introduction

While the work of dealing directly with site and subgrade problems is likely to be performed by consultants, the architect will retain the managerial functions and judgments that involve an overview of the scope and completeness of the work. In order to perform those tasks responsibly, architects must have a general understanding of the composition and structure of typical surface and near-surface materials and the methods to gather the necessary information about them. Architects must be familiar with the terminology and the basic concepts of soil mechanics as well as foundation engineering and construction. It is also important to possess an appreciation of the problems of working with ground materials for the purposes of building and site construction and general site-surface development.

Further, architects must understand how site and subsurface conditions can affect building and site design and vice versa. Lastly, it is essential for architects to have a familiarity with the general functions ordinarily provided by the other professionals (geotechnical specialists, civil engineers, structural engineers) who perform much of the work of discovery and evaluation and who then use the information to do the necessary design work for site and subgrade construction.

This mini-monograph will provide a basic understanding of aspects of the design and construction processes.

Basic Concerns

Basic Concerns

For architects, as building and site designers, the concern for soil and subsurface water conditions derives essentially from a need to accommodate them as they perform their design tasks. Areas of concern include the nature of the design tasks, the situation and context for the design work and the need for information to support responsible design. Before addressing site materials and engineering problems, architects should consider the basic needs for design work that establish the basis for knowledge about subsurface conditions.

Physical Design and Planning Tasks

Design tasks relate to subgrade conditions. As the site development progresses, an important consideration is the reshaping of the site surface with regard to cuts and fills. Use of the existing subsurface materials is involved in this process, with possibilities for exposure of lower deposits, use of cut materials for fills at other locations, establishment of slopes and the need to dispose of excessive or unwanted materials.

General development of the existing site as a functional property and as a location for plantings involves many considerations for use of the surface and near-surface materials on the site. Major factors are the use of irrigation, retention of existing plantings or other site objects and the general stability of the site in its natural state before site development.

Except in urban areas, where the building footprint may virtually cover the site, there will almost always be some site construction, usually involving paving, walls, retaining structures, various underground elements and bases for lighting and signs. Surface and near-surface ground materials must provide adequate support of these constructions.

Physical Design and Planning Tasks

Service elements for off-site supply of water, sewers, gas, electrical power, cable television and telephones often involve surface and below-grade construction. Installation, maintenance and modifications of these elements are affected by the subsurface conditions of the site.

Adequate support for the building is a major consideration for every site, whether it involves modest, near-surface footings or deep foundation elements of piles or caissons. Since the ideal situation is for a building to be supported on undisturbed ground, the exact knowledge of below-grade conditions is obviously a critical concern.

Many buildings are built with some spaces below finished grade level. Below-grade site conditions are critical for initial excavation, performance of construction work and for their effects on the finished building spaces for the lid of the building. Constructing below-grade spaces may be too difficult and expensive if excavation for construction involves rock cutting or blasting, dewatering or underpinning of adjacent properties. A groundwater level above the floor of below-grade space would require expensive water-repelling construction and could involve critical hydraulic pressures and uplift on the structure.

Physical Design and Planning Tasks

Water problems include surface water (runoff of precipitation or melting snow and ice), and irrigation arm groundwater. Potential problems are numerous and include concerns for erosion, stability of surface and near-surface ground materials and water damage to below-grade construction. Indeed, just about any modification of an existing site disturbs the natural equilibrium of both surface and groundwater conditions and can be a source for major readjustment of the ground materials. Design for these effects must deal with both existing conditions and the long-term effects of the destabilization caused by changes in the site conditions.

Individual sites may have various concerns for seismic activity related to site conditions. A general risk may exist for the geographical area relating to the history of earthquakes or to known, nearby faults. More specifically, the actual seismic properties of the site may be of concern: soil liquefaction (acting like a liquid) is a particular risk related to soil conditions. The choice of the building structure may relate to the potential for dynamic building or site interaction during seismic activity.

Site development may involve various situations that require extra consideration for existing site conditions. Some common situations include excessive reduction (cutting) of the site surface, which will expose ground materials previously buried and can seriously alter the surface drainage conditions. It also presents the problems of the removal and disposal of the cut materials.

Physical Design and Planning Tasks

Another provision may be excessive fill to raise the site surface level. This involves finding and importing the fill materials, properly and carefully controlling installation of fill to obtain a stable finished grade and providing adequate support for near-surface construction (pavements, site walls, shallow building footings, etc.).

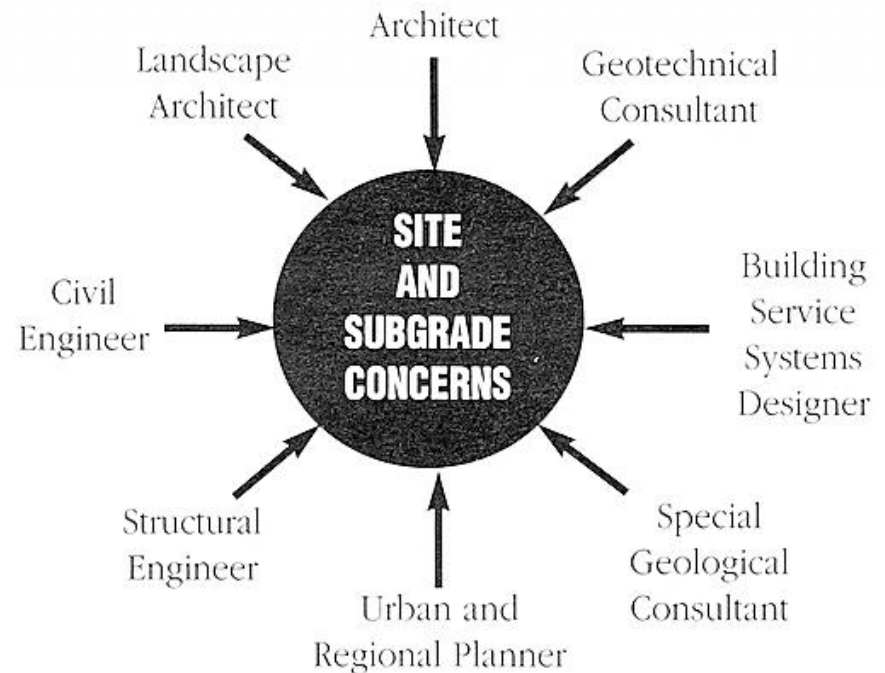
Another common situation, deep excavations, may be difficult to achieve in some ground materials (rock, for example) or because of near-surface groundwater. They may also present the problem of removal and disposal of excavated materials, if they cannot be used elsewhere on the site floor for fill. If excavations occur near the site boundaries, they may present concerns for the protection of adjacent properties, buildings and existing street or utilities.

There are of course many other possible design-related concerns, such as waterfront and hillside locations and redevelopment of sites with existing construction. Each building and its site present a unique situation, typically involving many commonly experienced conditions and perhaps a few special ones.

Design Coordination

Surface and below-grade considerations often involve many specialists in the design and construction work (see *Figure 1*). A major task for the architect as the prime design professional is to ensure the overall integration of the design work through coordination of the individual efforts by these various special consultants. As in many other situations, the architect must know when the advice or assistance of consultants is required and from what sources it must be obtained. For subgrade conditions, typical consultants are geotechnical specialists, structural engineers, foundation design specialists and civil engineers.

Figure 1 PARTICIPANTS IN SUBSURFACE CONCERNS



Design Coordination

Geotechnical specialists provide services for discovery of site conditions, analysis and evaluation of samples and tests, consultation for development of general design criteria, recommendations for the proper foundation system to use and recommendations for resolving potential site and subgrade problems.

Structural engineers are the design professionals who perform most of the ordinary design work for site construction and building foundations. Their work is often viewed as an extension of the design for the building's structural system.

Foundation design specialists may be structural engineers, geotechnical specialists, or both, who deal with a broad range of special foundation design problems. For various special types of excavation or foundation problems, they may be associated with an organization that specializes in foundation construction or with the manufacturer of a proprietary system, such as a particular form of pile or caisson.

Design Coordination

Within the field of civil engineering, there are specially qualified individuals and firms who deal with such site-development matters as surveying, roadways, water-handling systems, disposal systems and environmental hazards.

It is possible to simplify the architect's coordination tasks if all these functions are provided by a single organization. However, it is also possible that all of these consultants, plus other specialists, are separate entities and thus require considerable coordination by the architect.

Since many design-related tasks call for the expertise of more than one professional, a clear definition of the specific responsibilities of each professional must be defined prior to the start of the design work. If all likely professionals work on a single project (architect, landscape architect, civil engineer, geotechnical specialist and structural engineer), they will have overlapping responsibilities in solving problems and performing tasks. Unless the architect, as the overall manager of the project design, carefully orchestrates and directs these activities, work may be duplicated by different members of the design team, which would result in conflict and waste of effort. What is worse, some tasks may fall through the cracks, leaving certain design work uncompleted.

Design Coordination

Historically, building failures related to subsurface conditions have accounted for a high percentage of liability claims in the construction industry. Every professional is legally accountable for that portion of the work that he or she performs. This is a concern that will affect the division and assignment of design work to ensure the effective accomplishment of the project, as well as the appropriate legal accountability by the individual designers or design teams.

Assuming that the various components of the design work are appropriately assigned and that the work is properly executed by each professional, there remains the task of coordinating the various pieces of design work into a whole and harmonious design. This will be greatly facilitated if the separate design tasks are thoroughly defined, including emphasizing all points of necessary coordination by each team member with others. If that coordination occurs as a developing process throughout the design work, the integrative effort at the end of the design process will be much smoother.

As the number of participants increases—as is the case with much of the site-design activity—the design process becomes more complex. Thus, it is crucial to share all information that is obtained in the general discovery process regarding the site and subsurface conditions; the architect should fully inform all the affected participants.

Design Coordination

To resolve site and subsurface issues, various aspects of the design must be carefully scheduled within the overall building- and site-design processes. It is critical that the architect obtain timely information for design work, establish a framework for major decisions and facilitate a process that overlaps design and construction.

No serious design for site and subsurface work can be performed without applicable information when it is needed. This means that the discovery process should be started as early as possible. The most basic decisions for placing the building on a site, in the earliest preliminary design work, should be made with valid data concerning subsurface conditions.

Critical design decisions that affect the general building design or the progress of the work for the site and subgrade construction should be made in accordance with a firm, pre-established schedule. For example, finding out late in the design process that the building needs a deep foundation, that it is positioned badly for groundwater conditions, or that it otherwise does not fit well with the site conditions will hinder the progress of the design and the final cost of the project.

Design Coordination

For large building projects, it is common to begin the site and subgrade work while the final design is still being done for the building itself. It is critical that all pertinent design information be available as early as possible.

Case Study: Oozing Clay

Soft, plastic clay may be a rich source for brick-making, but it generally poses many problems as a subsurface material for construction. Building in downtown Chicago, for example, means dealing with a thick seam of soft clay that underlies most of the area, perched in a lower strata between a top crust of sand and fill and rock below.

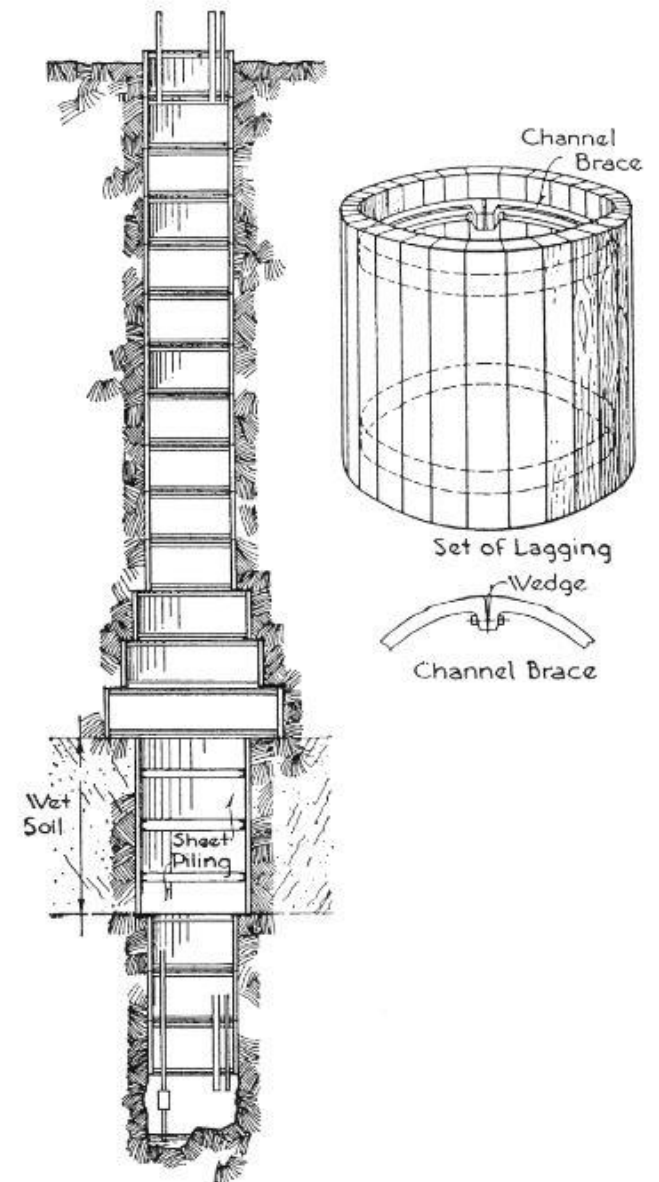
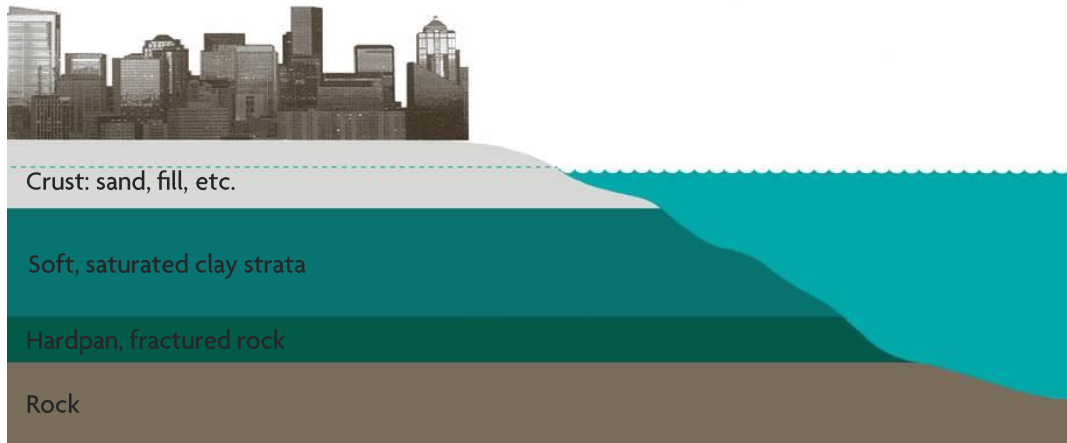
Small buildings without deep basements can sit on the crust at sufficient distance above the clay without being affected by it. However, with the advent of the skyscraper in Chicago came much heavier loads. Once outside pressure is delivered to the clay it will seek release, like toothpaste in a tube. This begins a steady oozing process and subsequent volumetric adjustment of the clay mass. The clay can ooze in any direction, including up, and can move objects in its path sideways or up. Light structures can still perch on the crust above the clay, but the only solution for the very heavy structure (high-rise building) is to penetrate the clay and seat piles or piers in the hard materials beneath the clay.

Case Study: Oozing Clay

The so-called "Chicago caisson" was developed to drive large piers down to the bedrock. The pier walls were lined to keep the pressure from being relieved in the clay, except for a small amount being excavated in the bottom of the shaft each day. This was, in some ways, analogous to the caisson method used for bridge pier excavations under water, which consisted of slowly lowering a box without a bottom by digging out the soil at the open bottom. With the Chicago caisson, the box did not move, it just grew longer downward and without the airlock.

This is an example of a soil condition that has a tenuous stability with the potential for major changes due to construction. Excessive cut or fill, change of groundwater conditions, adding stress of any kind or exposing the problem soil to air and precipitation may create difficulties.

Case Study: Oozing Clay



Design Information

Design Information

It is desirable, of course, to know as much as possible about the subsurface conditions for a proposed building site. However, the information that is explicitly needed for actual design work tends to focus on specific design tasks and actual elements of the construction. For purposes of understanding the need for specific information and the ways of obtaining it, consider three key questions:

1. What types of design tasks are required for the proposed building and site construction?
2. What specific information about subgrade conditions is needed for the design work required?
3. How can the necessary information be obtained? From whom? At what cost? With what reliability? With how much time allowance?

For illustration purposes in addressing these questions, consider a relatively simple task: the design of shallow bearing footings for a low-rise building.

Listing Design Tasks

A list of typical construction elements would be:

- Site-cast reinforced concrete elements for support of columns and bearing walls (footings)
- Basement walls and/or grade beam elements between the footings and the building construction above grade
- A concrete pavement slab (slab on grade) as the building first floor, whether as a basement or ground floor
- Site paving for walks, driveways, patios, etc.
- Other site construction as required: curbs, retaining structures, bases for signs and lighting, below-grade trenches, tunnels and vaults for utilities, etc.
- Installation of plantings: lawns, ground cover, flowers, shrubs, trees, permanent irrigation systems

Determining Information Requirements

Timing is an important consideration. Some information is desirable even before the decision to use the site is made, such as any previous soil borings, any previous construction on the site or the details of foundations of adjacent buildings. A different class of information is required before any concrete is poured for site or foundation construction. This type of information may be obtained by asking the following questions.

What is the general stability of the site soil masses? This is often the single most important issue. Is the general soil mass underlying the site likely to remain in its present form or might there be problems with erosion, surface settlement, downhill slip, liquefaction during seismic activities, frost heave, soil expansion, collapse of high-void soil, a sinkhole or other effects involving major readjustment of the soil mass?

Determining Information Requirements

Can excavations and deep cuts be easily achieved? Many factors are involved, but it is critically important to know the exact nature of the subsurface soil and the possible presence of a high groundwater level. If rock is present, the work for extensive rock excavation or blasting can be both difficult and costly.

For sites where extensive regrading is required can cut materials be used for fills in other locations on the site? If not, it may be necessary to import large volumes of useable fill materials.

Will disturbed and relocated soils be stable for all future situations? This includes concerns for dry and wet conditions, freeze-thaw cycles, vibrations and impacts, seismic movements and any other foreseeable phenomena.

Will major cuts or excavations disturb the existing subgrade soil masses in critical ways? This work can affect adjacent properties as well as the building site itself.

Determining Information Requirements

Can a stable site surface be established, especially for support of pavements, slab floors on grade and site construction? There are many factors to consider. Some may be controlled by design; some may be beyond control.

What value can be used for permissible maximum vertical pressure for footings? What values can be used for lateral effects: sliding friction, active horizontal pressure (how hard the soil pushes against retaining structures) and passive horizontal pressure (how hard it is to push something horizontally into the soil mass)? Discovery and testing are required to obtain these criteria. Usable soil property values for design will also affect the vertical positioning of footings; one answer may be that footings are simply not feasible and a deep foundation system is required.

What is a reasonably predictable amount of settlement of footings given the soil bearing capacity at the project site? Will differential or long-term settlements be a problem? The complexity of these issues depends on the building size, shape and structure.

Determining Information Requirements

What is the seasonal fluctuation of the groundwater level in the soil (water table, free water level etc.), and how does it affect the site soil masses? These concerns can critically affect the design for underground construction, the excavations or the stability of the soil masses in general.

Will freezing of the soil be a problem? If so, what is the anticipated depth of frost? These basic questions identify issues that can affect most building and site designs. Many more questions apply to special design situations-such as high-rise or long-span construction, high seismic risk or special site conditions such as hillsides or waterfronts.

Information that answers some of these questions may be obtained from local or government agencies or from other property owners. However, the reliability, of such information should always be verified through actual project-specific discovery. General stability of existing soil masses and groundwater conditions can probably be established, however, from these sources. The geology departments of local colleges and universities can be helpful in locating sources of such information.

Determining Information Requirements

Most useable design criteria, however, must be established with some discovery process, typically involving exploratory activities on the site. Design decisions must depend upon a combination of observed and measured properties.

Soil profiles are established by deep soil sampling and involve the identification of the basic materials and their vertical positions in the typical layered soil mass. When layers are not consistently uniform or horizontal, a considerable number of explorations may be required for a large site.

Basic properties for each soil layer of concern include values for the in-place soils such as unit density (weight), water content, grain size and shape and range of graded sizes in the soil mixture at a single level.

Basic tested values include dry unit density (of oven-baked samples); effects of water content variation (particularly for silts and clays); penetration resistance for sands (observed while extracting soil samples); unconfined compression resistance (for clays) and general permeability of the soil mass (resistance to water flow).

Determining Information Requirements

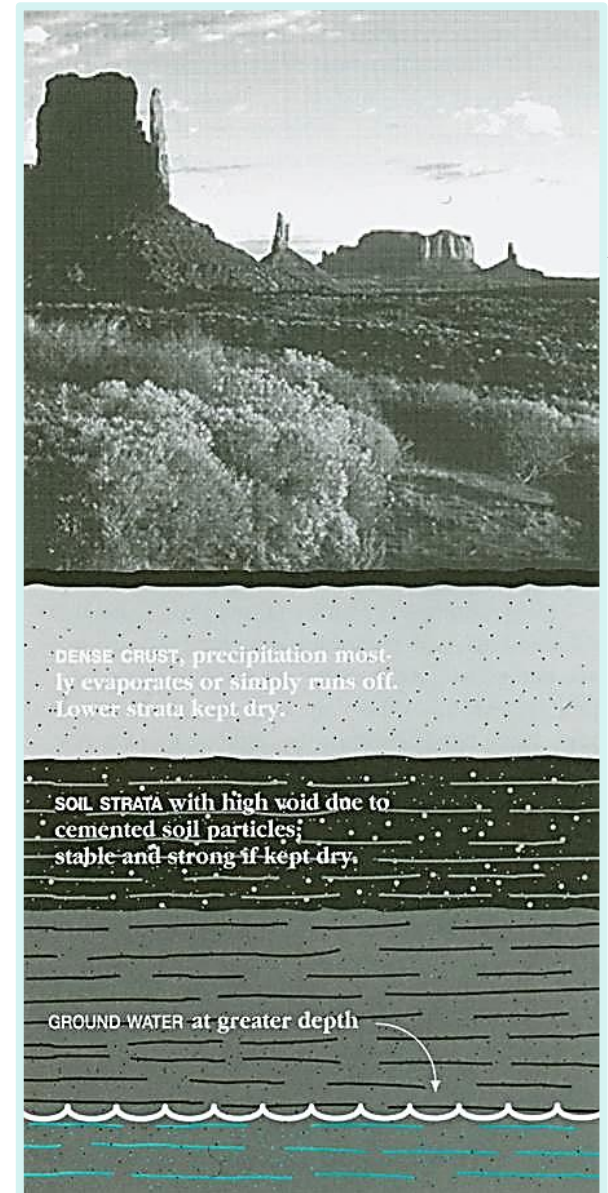
Case Study: Dry Soil, High Void

A situation that occurs in areas with arid climates is one in which natural processes have created a soil deposit with a high void, with the void developed through the cementing of soil particles into a series of tiny caverns. Fine soil mixed with water can produce a fairly strong natural concrete-like material when dried (adobe brick, for example), and the soil may have considerable strength and deformation resistance if kept dry. Centuries may go by as the climate sustains the condition.

When new communities are built, people want grass in the desert, or maybe an unlined pond for a fountain, a swimming hole or a water reservoir. This causes water to seep steadily down into the voided soil, dissolving the cemented walls of the voids, and a major consolidation occurs. The ground surface sinks along with anything sitting on it.

Determining Information Requirements

Superficially, one solution may include using artificial turf. With present technologies, it may be possible to line the pond or reservoir with a geotechnical fabric to retain water from the soil below. A more permanent and reliable solution is the preconsolidation of the soil by flooding the site to cause the collapse before construction begins. If the problem soil deposit is relatively small (thin strata), it may be possible to modify it by intrusion of an agent to fill the high void and significantly stabilize the soil structure. If the problem soil deposit is close to the surface, it may be possible to dig out the voided soil and replace it with a compacted one. Of course, the best solution may be simply to develop the site with more sensitivity to natural conditions that recognize the situation of the arid climate.



Obtaining Information

As discussed previously, some information of record may be obtained easily with little or no cost to the user. For modest projects, in areas with considerable construction experience, such information may suffice for actual final design. Most building codes (and enforcing agencies) allow for some design work to be done with so-called presumptive values, based on prior experience.

For larger projects or special site or subgrade conditions, however, more information—and possibly some expert consultation—is often required. Deep soil sampling, testing of soil materials and extrapolating useable design values will typically be done by professional organizations specializing in geotechnical services.

Obtaining Information

For design work that relates to subgrade conditions, there are many sources of information and types of information. A critical part of managing the design process is the establishment of a schedule for gathering and utilizing this information. Common sources of information about subgrade conditions include previous local work, government agencies, geotechnical discovery programs, “dig and learn,” post-construction monitoring and the type of information needed for design.

Building construction, highway construction or other activities in the immediate vicinity, or maybe even on the actual building site, will have generated information, both as pre-construction discovery efforts and as real situations encountered during construction. Any post-construction difficulties that may have occurred are especially important. Information gained from construction experience may be obtained from design firms, contractors, owners or local media archives. Of course, geotechnical specialists in the area will have working files containing this kind of information.

Obtaining Information

Many different government agencies, from local to federal levels, have information for both site surface and subsurface conditions. Some of these are quite general and others are very specific as to purpose. Many agencies—highway, agricultural, water management, mineral resource management and military—have vast resources of information. Much of this information is free and readily accessible. If architects don't ask for it, geotechnical or structural consultants will, and they may charge for it. It pays to learn about all these sources and about how to use them for preliminary design information.

The most reliable information is likely to come from a planned program of discovery activities developed specifically in relation to the design work and the project work schedule. Developing this discovery program is the first thing that any geotechnical consultant should undertake. It should be noted that a preconstruction discovery program is designed to develop an understanding of typical subsurface conditions, and it will not usually be extensive enough—because of cost and time restraints—to identify all variations in the subsurface conditions.

Obtaining Information

The most thorough pre-construction discovery effort may not fully reveal what digging on the site will actually encounter. Experienced designers, consultants and contractors know this and try to provide for it to the degree that they can. Certain discoveries are anticipated and are incorporated in any planned discovery process, with possible timely inspection of the construction work and alterations of the designed work anticipated as well.

During the early life of the construction, special problems may occur. Therefore, a monitoring program should be set up to observe the actions of the ground, site surfaces and structures for some period after construction. Monitoring may be required because of ground conditions, the form of the construction, the magnitudes of supported loads or the sensitivity of occupancy conditions to movements. If any post-occupancy remedial efforts are necessary—such as drainage, soil stabilization, underpinning, jacking, etc.—it is best to discover the need before it becomes a serious, damaging condition. And in the interest of a healthy designer-client relationship, it is best to communicate the possibility of such a condition early on.

Obtaining Information

All design work occurs in successive stages, and at each stage the assurance of the correctness of the work is conditioned by the degree of accuracy or completeness of the information available at the time of design. Thus, some preliminary design work may be informed only by pre-existing information, before any project-specific discovery work is undertaken. However, the final construction should be done with design work that is informed by a well-planned discovery program.

For any specific design task it is easy to describe the type of information architects would like to have in order to feel secure about the design. However, for much of the subsurface construction work, it is usually necessary to settle for what can be logically and feasibly obtained at the time the design must be accomplished. It is often necessary to proceed with the completion of the construction documents, the bidding process, negotiation of contracts and the beginning of excavation and grading work with what amounts to preliminary information in the full discovery process.

Obtaining Information

At any point in the design process, whatever information is available must be evaluated for practical use in the design work. The designer may have simple needs for answers to questions relating to the maximum usable bearing pressure for the design of footings, the predictable settlement due to vertical loads, the anticipated horizontal pressure on retaining structures and the predictable highest level for groundwater. These are relevant concerns, but getting answers, especially in early stages of the discovery process, is not likely to be easy—at least if some real assurance of reliability is expected. There are likely to be problems. For instance, discovery data doesn't always translate easily into design criteria. Discovery operations and lab tests tend to produce forms of data that cannot be applied directly to the design process. In many cases, combinations of data must be interpreted to extrapolate a practical design value. Typically, a range of possible values is indicated, and picking a specific one is at least partly judgmental.

Design processes may not relate well to the specific problems of the site and subsurface conditions. Site planning, building placement, choices of construction materials and forms and other early basic decisions may not be made with full knowledge of the real situations revealed by subsequent discovery or during the actual construction work. This may put the designer in the position of trying to design something that is essentially wrong for the conditions.

Question

In order to obtain and understand the information they need to create a responsible design and understand it properly, designers must know which questions to ask. List the critical questions that a designer should ask during the design phase.

Answer

In order to obtain and understand the information they need to create a responsible design and understand it properly, designers must know which questions to ask. List the critical questions that a designer should ask during the design phase.

Designers should ask questions such as:

- What is the general stability of the soil?
- Can excavations and deep cuts be easily achieved?
- Can cut materials be used for fill elsewhere on-site?
- Will disturbed and relocated soils be stable for all future situations?
- What values can be used for maximum vertical pressure for footings, lateral effect, and passive horizontal pressure?
- What is the predictable amount of settlement of footings?
- Can a stable site surface be established?
- What is the seasonal fluctuation of the groundwater level?
- Will the soil freeze, and if so, to what depth?

Division of Responsibilities

Division of Responsibilities

A major objective of any architectural project is to ensure that all the necessary design work is performed in a competent and responsible manner. The overall responsibility for this rests with the architect as the prime design professional. However, specific design tasks are often performed by others, possibly because of the size and complexity of the project, or because of the need for a level of expertise that is outside the range of architectural practice.

No matter how many participants there may be in a project's design, it still remains a major task to ensure that the work is complete and responsibly done. For each project and task, it must be determined if specific tasks will be assigned to others or if the architect will complete them. Some considerations regarding subsurface conditions are the size and nature of the project, the complexity of the subsurface problems and the available consultants.

Size and Nature of the Project

For practical purposes, the design work for large projects must be divided among many people. The logical way to do this is to take parts of the design work that represent established professional specialties and to assign them to those specialists, whether employees of the architect or independent consultants.

For almost every aspect of the site and subsurface work there is an available specialist, making it possible for whomever is in charge to assign almost all the design work to others. For large projects, it may be both wise and practical to do this, enabling the architect to coordinate the work and integrate it with the rest of the project design work.

However, the smaller the project, and the more routine the work, the more likely it is that the architect may do a larger share of the design work. Alternately, an individual architect or firm may have the capability and experience in some special phase (such as site design or foundation design) and be inclined to do the work without consulting outside professionals, no matter how large the project.

Complexity of the Subsurface Problems

A major responsibility for the architect is to determine when it is necessary to seek expert advice. Design for ordinary subsurface work can be done with a great deal of reliance on precedent. For project types that are done repeatedly, most architects eventually acquire the confidence to do increasingly larger portions of the design work themselves in order to save time.

In managing the whole design effort, there is hardly a more critical judgment for an architect to make than when to call in the experts. The first step is to judge whether the situation is “routine” or whether there are indications of potential problems. This obviously requires some knowledge of what makes a situation routine or special; the lower the level of knowledge, the more an architect should seek help—even in deciding whether help is needed. Even experienced architects may find themselves confronted by new conditions.

Answering these questions should give some idea of the relative need for help.

Complexity of the Subsurface Problems

Has there been any previous construction on the site or in the immediate vicinity with similar conditions? If so, inspection of the construction or conversations with the owners, builders, or designers can be enlightening.

Is there any evidence of major distress to the site or to the immediate vicinity? Learn to recognize the evidence of major erosion, landslides, sink holes, seismic faults, flooding and other events that indicate potential problems.

Is there any available information of any kind about subsurface conditions? Try the various potential sources described previously in the discussion on design information. It is hard to imagine a locality where previous work has not been done with some discovery information.

Is the nature of the proposed project one for which subsurface conditions are critical? This question prompts others that must be considered for each project: Is there to be extensive underground construction? Will loads on foundations be great? Is settlement of supports a critical problem for the type of structure or the building materials (plaster, masonry, etc.)?

Available Consultants

Once it has been determined that help is necessary, the next step concerns the need for some evaluation of the various resources and the specific scope of their services. Consultants' fields overlap, and some may offer recommendations of a broader nature than you request. This might lead to other consultants or to a package deal for multiple consulting services from a single source. There are several potential sources you can go to for assistance.

Other architects, who are more experienced, or who have had prior experience with the specific conditions you are facing, can be very helpful. They may not eliminate the need for additional specialists, but they can look at the problems from the same point of view.

Available Consultants

Civil engineers are generally the design professionals most involved in site and subsurface problems. Their firms often provide all the services for the site and subsurface work, from surveying to complete design of foundations and site construction. They also do design work for site grading, water control, traffic management and other site engineering problems. If the major problems you anticipate are structural ones, structural engineers are the experts. They may be specifically identified as such, or they may also be civil engineers.

Although they vary somewhat in the scope of services they offer, landscape architects combine the work of architects, site planners and civil engineers as necessary for full site development.

Geotechnical specialists, once known as “soils consultants,” are often associated with organizations that provide a range of services including site exploration (soil borings and field tests), lab testing of soil samples, general consultation on subsurface conditions, provision of inspection and testing during construction and possibly complete design of site and foundation construction.

Question

Architects need to accommodate soil and subsurface water conditions during the design phases of every project in order to create a responsible design. To do this, they must create as comprehensive a survey of these conditions as possible, as early as possible. Describe the various ways and sources an architect might use to access this required subsurface information.

Answer

Architects need to accommodate soil and subsurface water conditions during the design phases of every project in order to create a responsible design. To do this, they must create as comprehensive a survey of these conditions as possible, as early as possible. Describe the various ways and sources an architect might use to access this required subsurface information.

There are two broad categories of information sources. The first is the more general information about local soil and water conditions, and the other is the more specific information about the site itself.

General information may be obtained from such sources as previous local building and highway construction work; government agencies such as highway, agricultural, water management, mineral resource management, and military agencies; geotechnical discovery programs; other design firms; contractors; and/or local media archives.

Site-specific conditions can be obtained from previous on-site soil borings or construction, or the details of foundations of adjacent buildings. They may also be provided by geotechnical specialists who can conduct soil tests and obtain samples to establish the design criteria for the conditions identified and make recommendations for the proper foundation system as well as for resolving potential site and subgrade problems.

Glossary

Glossary

Active lateral soil pressure

The horizontally directed pressure exerted by a soil mass against a retaining structure (retaining wall, basement wall, etc.).

Batter pile

Pile driven at an angle to the vertical to develop significant lateral bracing.

Bearing foundations

Foundations that resolve loads through vertical pressures spread over a supporting soil surface. Also called spread footings. If occurring near the ground surface they are also called shallow bearing foundations.

Bonded soil

A soil in which the large particles are held together by some action, so that the soil mass has a general resistance to deformation.

Glossary

Caisson

Common name for a pier. Technically, a caisson is a device used to advance the excavation for a large pier, usually underwater.

Cemented soil

A bonded soil in which the bonding action consists of a cementing action caused by the finest soil particles. May usually be loosened by water, unless there is some significant chemical bonding, as with portland cement.

Clay

A common type of soil, usually consisting of the finest size particles in a soil mass.

Compaction

Action of compacting (packing tighter) surface soils to produce denser supporting materials, usually for paving or for slab floors on grade.

Glossary

Consolidation

Reduction of the void (open space) within a soil mass, resulting in a closer packing of the solid particles and an increase in the soil unit density (weight per unit volume).

Cut

Removal of existing surface ground materials.

Deep foundation

Foundation elements extended to deep soil layers. Common forms are piles or piers.

Dry soil weight

Unit weight of a dried sample, determined by baking a sample in an oven.

Fill

Generally refers to “man-made fill,” consisting of a soil mass created by construction activity, rather than by natural processes; building up of the existing ground surface. Also used to describe the deposited materials used, as a soil type.

Glossary

Fundamental period of the site

Harmonic response property of a site soil mass, significant for evaluation of the potential building/site interaction during an earthquake.

Geotechnical consultant

Specialist in soils, subgrade problems and matters relating to design and construction of foundations and other structures on soils.

Geotechnical report

A report, typically prepared by a geotechnical consultant, describing the results and recommendations forthcoming from a discovery effort for subgrade conditions. Commonly called a Soils Report.

Glossary

Gradation of soil particle size

Breakdown of the sizes of particles in a soil sample by percentages in various size categories. Used to identify the general nature of the soil and its constituent materials. Basic types of soil materials are essentially identified by the size of their particles: from gravel to sand to silt to clay, for example. Generally established by *sieve analysis and sedimentation test*.

Grade

The top of the ground; called natural grade, existing grade or original grade for the existing site and finished grade for the redeveloped site.

Grade beam

A grade level element used to span across two or more isolated footings or over uneven bearing materials to provide continuous support for a structure.

Gravel

Basic soil type identified by the large size of its particles (larger than sand).

Glossary

Groundwater level

Level of free water in the ground. Determined most reliably by digging a hole and observing the level of standing water in the hole. Soil below this level will ordinarily be saturated with water. Also commonly referred to as the *water table*.

Loam

Soils with high potential for sustaining plant growth. Typically slow-draining (water-retaining), relatively easily broken up (pulverized) and high in organic content (mostly decayed plant materials). Also referred to as *topsoil* when occurring as the *topmost* soil layer.

Passive lateral soil pressure

Horizontally directed soil pressure developed by a soil mass in resisting the pushing of some object through it.

Glossary

Penetration resistance

Resistance to the advancing of a standard soil sampling device into the soil; measured by the number of blows required to advance the sampler one foot. An important indicator of relative density and strength of sandy soils.

Percolation test

Consists of digging a hole, filling it with water and observing the rate at which the water soaks into the soil. Basic measurement of the property of permeability of the soil.

Permeability

Relative ease with which water soaks into or drains out of a soil mass.

Pier

A deep foundation element consisting of a concrete column cast in an excavated shaft in the ground. Also commonly called a caisson. Belled piers have their lower ends flared out to work as bearing footings. Socketed piers are drilled into bedrock (socketed) at their lower ends.

Glossary

Pile

A deep foundation element, advanced into the ground by dynamic driving (pounding) or by vibration, for resistance to some combination of vertical downward force, upward force (called uplift) or lateral force.

A friction pile is driven until its length develops sufficient skin friction in the soil to support its load, or until it becomes too hard to drive. A special problem with friction piles is negative friction, in which the consolidation of soils surrounding the pile causes it to be dragged down after initial driving.

An end-bearing pile is driven a specific length to seat its end in rock or in some very hard soil strata.

Resistance of piles to loads is usually determined by the number of blows per foot needed to advance the pile just before driving is stopped.

Glossary

Plasticity

The numerical difference between two tested, water-related properties of a fine soil (silt or clay). High plasticity indicates a clay; low plasticity indicates a silt. Exceptionally high plasticity indicates a very plastic (soft, oozing, probably organic) clay.

Preconsolidation

Effect on existing soils, consisting of inducing volume reduction (consolidation) prior to construction, usually to reduce anticipated settlements of supported structures or subsidence of the ground surface.

Retaining device

Element used to resist lateral (horizontal) soil pressure. Includes basement walls, curbs, cantilever retaining walls for vertical cuts and various slope-retaining devices.

Glossary

Rock

Ground material generally identified by the difficulty in achieving its excavation (compared at the extreme to loose soil).

Runoff

Drainage from the site surface of water from precipitation, melting snow or ice or excessive irrigation.

Sand

Basic soil type identified by the relatively large size of its particles: smaller than gravel, but larger than silt and clay.

Saturated soil weight

Unit weight of a soil predicated on the complete filling of the void with water.

Glossary

Sedimentation test

Test performed on the finest soil particles (called the fines) remaining on the finest mesh sieve in a sieve analysis of a soil sample for soil size gradation. The fines are placed in a jar of water, the jar is shaken and the rate at which the particles settle to the bottom of the jar is observed. The coarser particles (silts) will settle in minutes; the finest particles (clays) may take days.

Sheetpile

Very thick steel “sheet” (plate) formed for stiffness (pleated, etc.), and driven into soil in interlocking sets to develop a retaining structure.

Sieve analysis

Test performed to establish the soil particle gradation in a soil sample, consisting of passing the materials through a series of successively finer wire-mesh sieves.

Glossary

Silt

Basic soil type with a generally fine particle size overlapping that of fine sand and clay. Distinguished from clay mostly by its water-related behavior (low plasticity compared to clays).

Sliding friction (coefficient)

Resistance to slipping (usually horizontally) at the face between soil and some structure (bottom of a footing, etc.). Predicted by a coefficient of friction for sands and gravels; by a unit resistance per square foot of the contact face for clays.

Slope angle

Measurement of the steepness of a slope. A critical value for design is the limit of the angle for a particular soil that can be achieved with some assurance of safety against failure of the slope.

Slurry wall

A “wall” of saturated clay (slurry) deposited in a trench to retain water or lateral soil pressures.

Glossary

Soil

General term for the materials of a loose nature (discrete particles) that constitute the ground mass.

Soil liquefaction

A unique response of a soil to dynamic vibration (as with an earthquake) consisting of turning a usually dry, firm soil almost instantly into a near-liquid state.

Soil strata

A layer of soil of a generally consistent character, clearly identifiable and distinguished from other layers (strata) above and below it in the general soil mass.

Stability of a soil mass

The general ability of a soil mass to retain its present nature, particularly with reference to its structural character.

Glossary

Subsidence

Sinking of the site surface.

Tieback

An anchoring device, extended into soil behind a retaining wall or slope-retaining structure, to develop more positive lateral resistance.

Till

Topsoil, occurring naturally or specially prepared for optimal sustaining of plant growth. Yields to tilling (plowing, etc.) to loosen and aerate the soil.

Topsoil

The topmost, ground surface soil materials. Usually implies *loam* or other material highly supportive of plant growth.

Glossary

Unconfined compression strength

Compression resistance of a cohesive soil (with relatively high clay content), consisting of crushing of a freestanding (unconfined) sample. Important for evaluation of strength of clays.

Unit density (weight)

Weight of a unit (cubic foot, etc.) of a soil. Three different weights are used for different purposes: weight as found in the soil deposit, dry weight as determined by oven drying of a soil sample and weight when saturated with water as determined by computation for the tested sample.

Uplift

Vertically directed, upward effect on a structure.

Void

Open space within a soil, not taken up by solid soil particles. May be filled with air or water.

Water table

See Groundwater Level

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Conclusion



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